

HARRISON 20 SSE, NE, USA

WMO: 744380

Lat: 42.4247N

Lon: 103.7363W

Elev: 1343

StdP: 86.20

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 94077

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-24.2	-20.3	-27.4	0.4	-24.5	-23.4	0.5	-20.1	9.4	-5.3	8.4	-3.8	1.4	N/A	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	33.9	16.7	32.2	16.5	30.5	16.5	19.8	27.8	19.0	27.3	18.2	26.7	3.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	14.5	22.9	16.4	13.8	22.0	15.6	13.0	21.4	62.8	27.4	60.1	27.3	57.1	26.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	6.3	-30.9	37.1	3.0	1.5	-33.0	38.1	-34.7	39.0	-36.4	39.8	-38.6	40.9		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-4.2	-3.7	2.6	6.2	11.1	17.5	21.2	19.5	14.5	7.1	0.8	-5.1		
	DBStd	10.45	6.35	6.77	5.71	4.90	4.78	3.85	3.05	3.24	4.29	5.21	5.74	6.76		
	HDD10.0	2106	441	383	233	129	45	1	0	0	9	117	278	468		
	HDD18.3	4217	699	617	487	363	227	59	10	26	127	349	527	727		
	CDD10.0	1144	0	0	5	17	79	227	348	295	146	26	2	0		
	CDD18.3	213	0	0	0	0	3	35	99	63	13	0	0	0		
	CDH23.3	4021	0	0	4	24	137	667	1508	1122	496	61	1	0		
	CDH26.7	1636	0	0	0	3	33	248	694	473	175	10	0	0		
(14) Wind	WSAvg	2.7	2.6	2.7	3.0	3.4	3.2	2.8	2.4	2.3	2.4	2.6	2.5	2.5		
(15) Precipitation	PrecAvg	370	6	9	18	42	61	67	46	37	33	30	11	7		
	PrecMax	506	15	30	52	85	152	149	77	73	83	69	40	20		
	PrecMin	207	0	0	3	5	20	15	15	10	5	2	1	0		
	PrecStd	72	4	6	12	22	34	40	18	16	19	18	9	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	16.9	23.7	25.9	30.0	35.0	36.7	35.5	32.9	28.3	21.9	15.3		
		MCWB	4.8	6.9	9.5	10.8	14.1	16.5	16.6	16.4	16.2	12.7	9.0	6.0		
	2%	DB	11.0	13.1	20.2	23.2	27.1	31.8	34.3	33.1	30.8	25.0	18.5	11.6		
		MCWB	4.0	5.1	8.0	10.7	13.6	16.3	17.5	16.2	15.2	11.5	7.7	4.2		
	5%	DB	8.2	10.0	17.1	20.2	24.4	29.7	32.7	31.5	28.8	21.9	15.2	8.7		
		MCWB	2.8	3.6	6.9	9.3	13.6	16.5	17.3	16.4	14.0	10.6	6.4	2.6		
	10%	DB	5.7	6.7	14.0	17.2	21.6	27.3	30.7	29.5	26.4	18.7	11.8	5.7		
		MCWB	1.5	1.7	5.8	8.1	12.7	16.2	17.5	16.6	13.8	9.2	4.9	1.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	7.9	10.8	12.7	16.9	20.1	20.7	20.3	19.1	14.0	9.5	6.3		
		MCDB	12.1	17.2	22.3	22.0	24.6	28.0	28.8	27.5	28.9	24.4	20.4	14.8		
	2%	WB	4.2	5.6	8.9	11.1	15.5	19.1	20.0	19.3	17.1	12.7	7.9	4.7		
		MCDB	9.7	13.3	19.6	20.5	22.5	26.5	28.2	27.3	25.9	21.5	17.5	11.7		
	5%	WB	2.9	4.3	7.5	9.6	14.4	18.0	19.3	18.4	15.9	11.3	6.5	2.9		
		MCDB	7.7	10.5	17.0	18.9	21.1	25.9	27.7	26.8	24.7	19.5	14.5	8.4		
	10%	WB	1.5	2.3	6.4	8.3	13.2	17.1	18.6	17.6	15.0	10.0	5.1	1.3		
		MCDB	5.5	7.3	14.3	16.2	19.4	25.3	27.3	25.9	23.7	17.8	11.8	5.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	14.5	15.8	15.2	15.0	16.5	17.0	17.9	19.2	17.8	17.1	15.6		
		MCDBR	18.2	18.4	21.8	22.3	20.6	20.2	20.6	21.5	24.0	24.6	23.4	19.3		
	5% WB	MCWBR	12.1	10.4	11.4	11.0	8.7	7.7	6.3	7.7	9.9	11.9	13.1	12.5		
		MCWBR	17.6	17.4	20.8	20.4	15.7	16.9	16.6	17.0	19.1	20.3	22.0	18.7		
(40) Clear-Sky Solar Irradiance	taub	taub	0.245	0.254	0.275	0.304	0.324	0.333	0.361	0.353	0.317	0.279	0.257	0.246		
		taud	2.591	2.559	2.532	2.451	2.464	2.460	2.415	2.419	2.507	2.588	2.566	2.568		
	Ebn at Noon	Ebn at Noon	934	973	981	966	948	936	905	903	920	928	905	900		
		Edh at Noon	64	78	91	107	109	110	114	110	93	75	65	60		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.15	3.02	4.25	5.25	6.12	7.26	7.21	6.25	5.02	3.41	2.36	1.80		
	RadStd	RadStd	0.10	0.18	0.32	0.30	0.50	0.44	0.27	0.21	0.30	0.30	0.14	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page